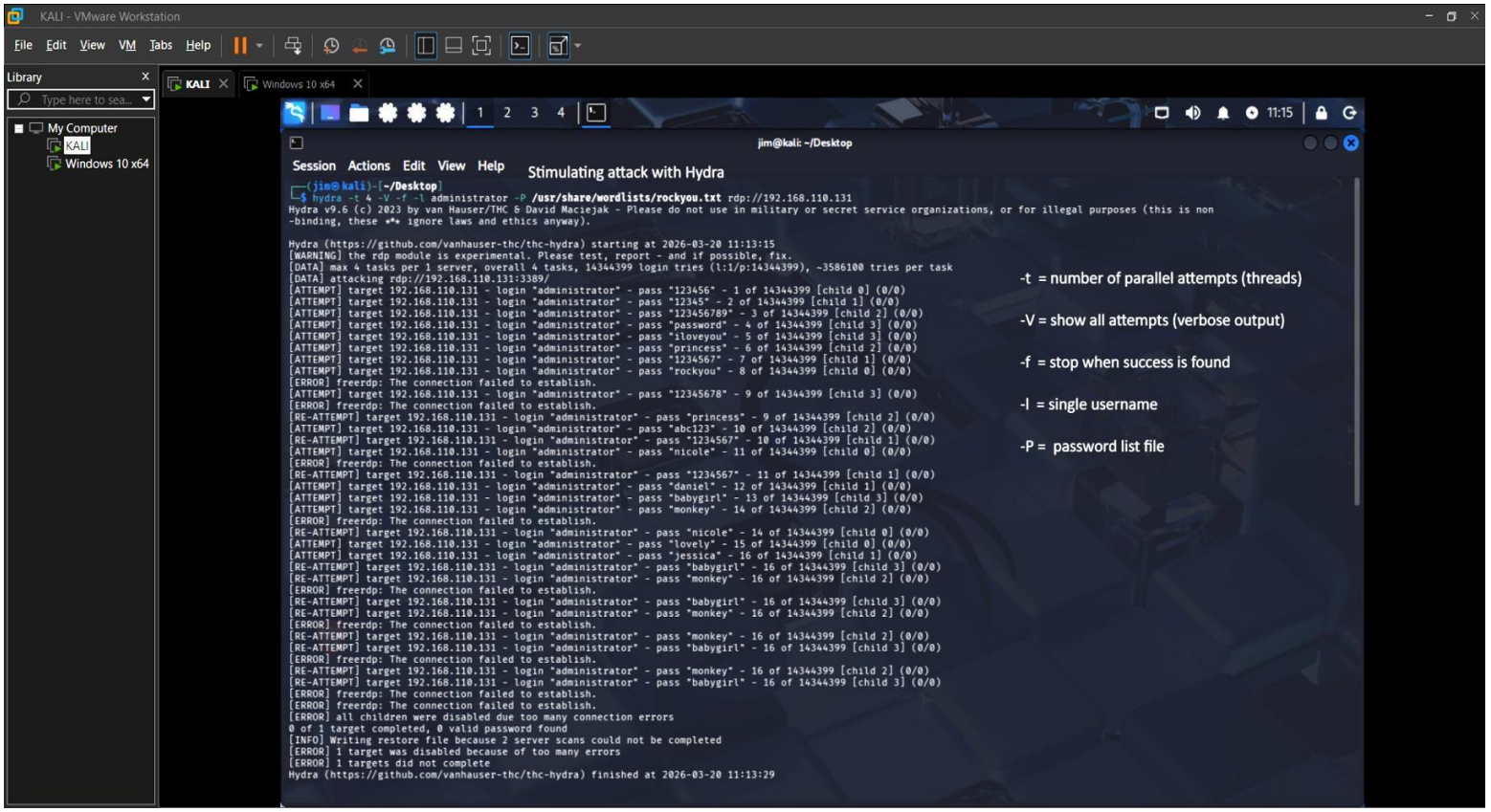
SOC INCIDENT REPORT
RDP Brute Force Detection using Splunk

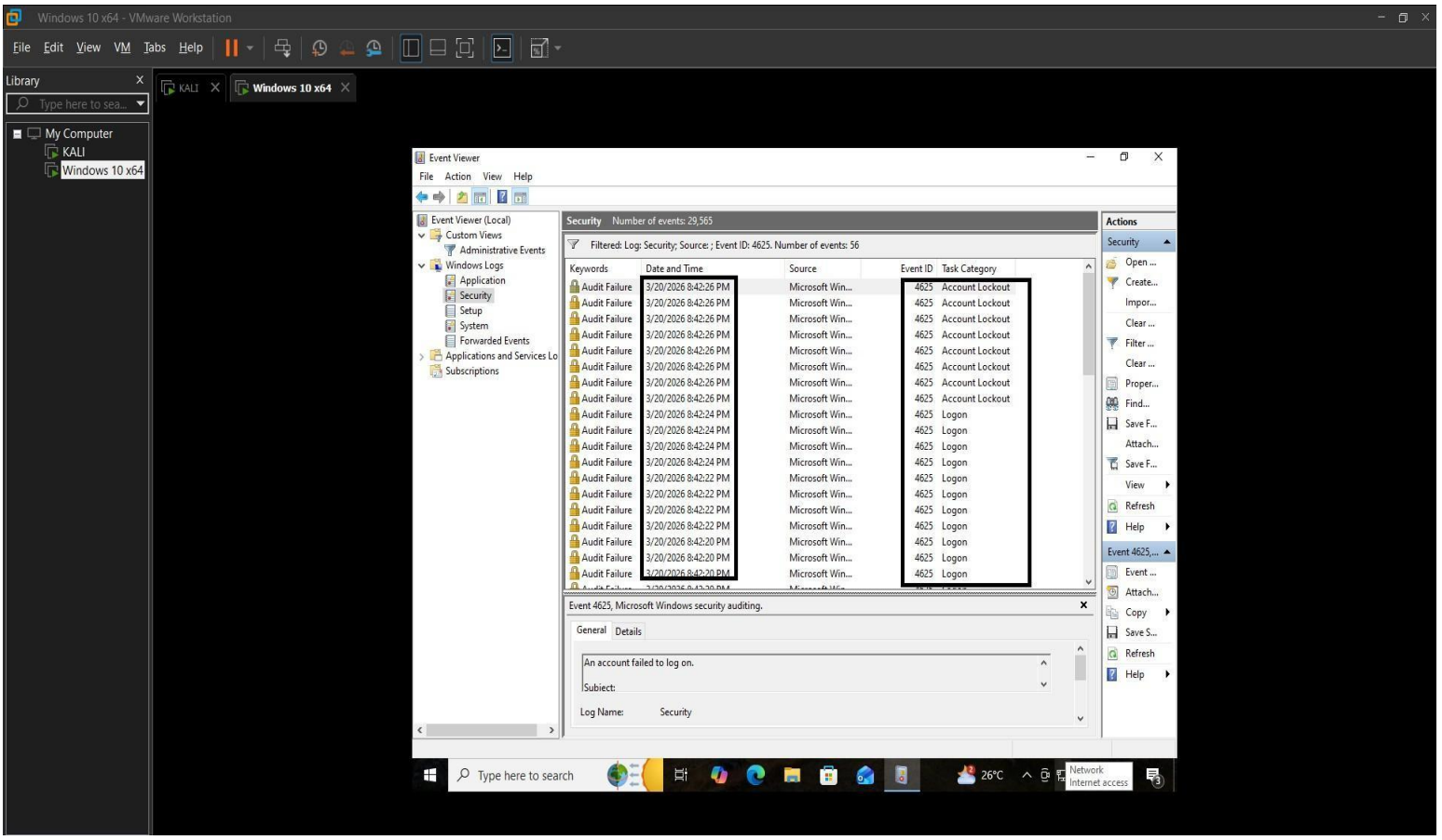
Objective:
Detect brute-force login attempts using SIEM by analyzing authentication logs.

Attack Flow Overview:
Brute-force attack initiated from Kali Linux targeting Windows via RDP (Port 3389). Failed login events are generated and forwarded to Splunk for monitoring and detection.



LOG ANALYSIS & DETECTION

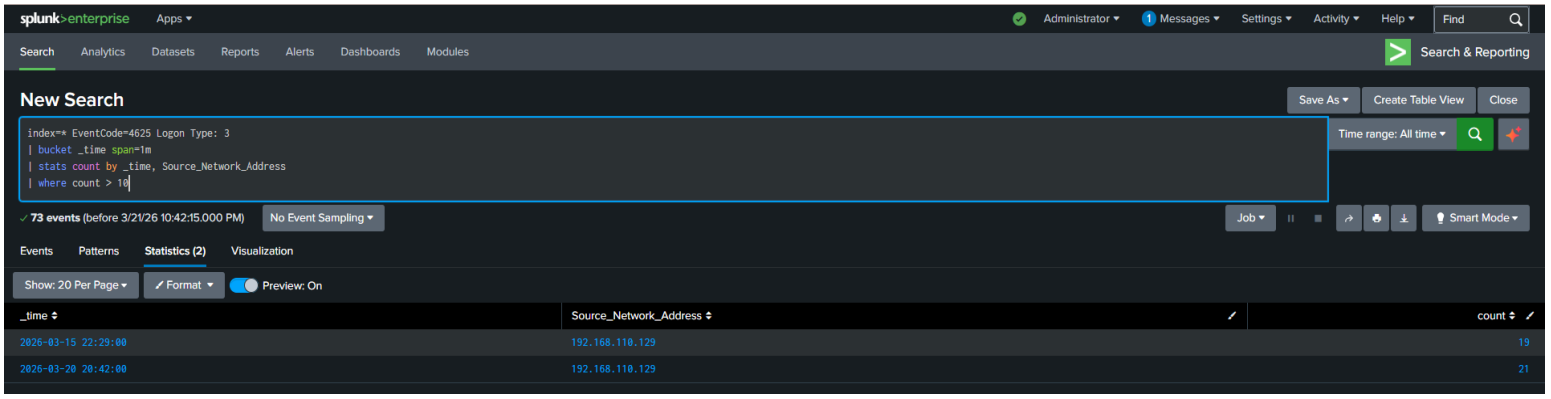
Windows Security logs recorded failed login attempts using Event ID 4625, indicating unsuccessful authentication attempts.



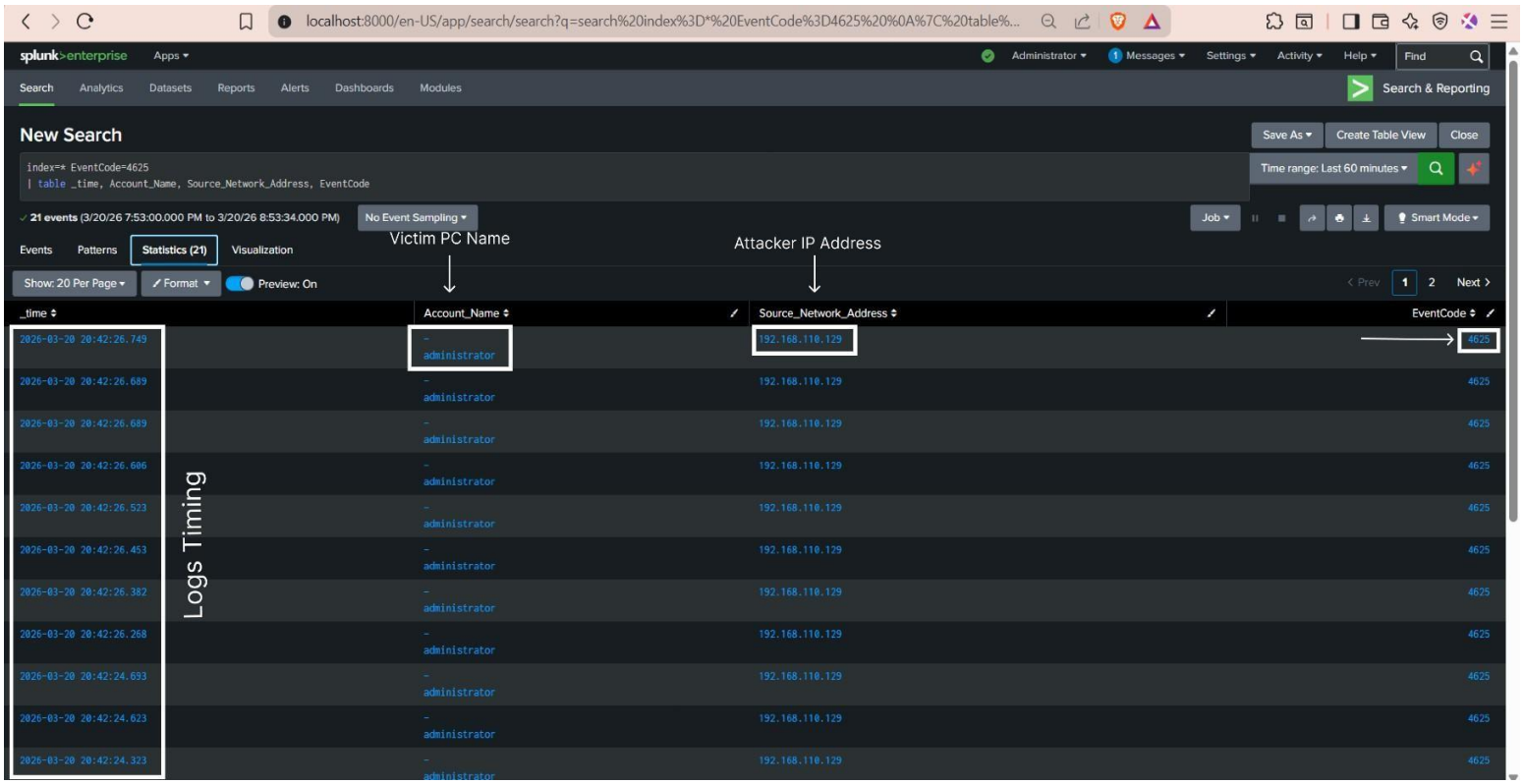
Detection Rule:

```
index=* EventCode=4625 Logon_Type=3
| bucket _time span=1m
| stats count by _time, Source_Network_Address
| where count > 10
```

Logon Type 3 indicates network-based authentication attempts.



INCIDENT INVESTIGATION



Logs were forwarded to Splunk SIEM and analyzed.

Timeline:

- Initial attempts observed at 8:42 PM
- Rapid increase within seconds
- Same source IP repeatedly observed

Observation:

High-frequency authentication failures from a single source confirm automated brute-force activity.

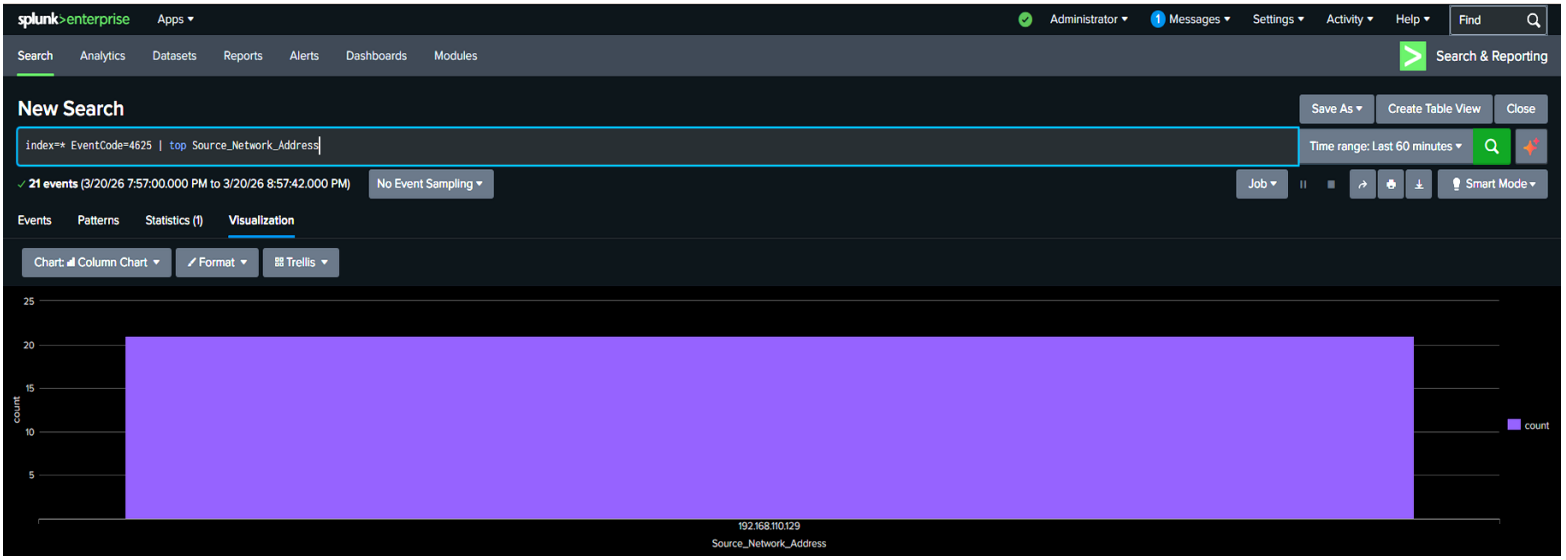
Incident Details:

- Source IP: 192.168.110.129
- Victim IP: 192.168.110.131
- Target User: administrator
- Event ID: 4625
- Logon Type: 3 (Network Logon)

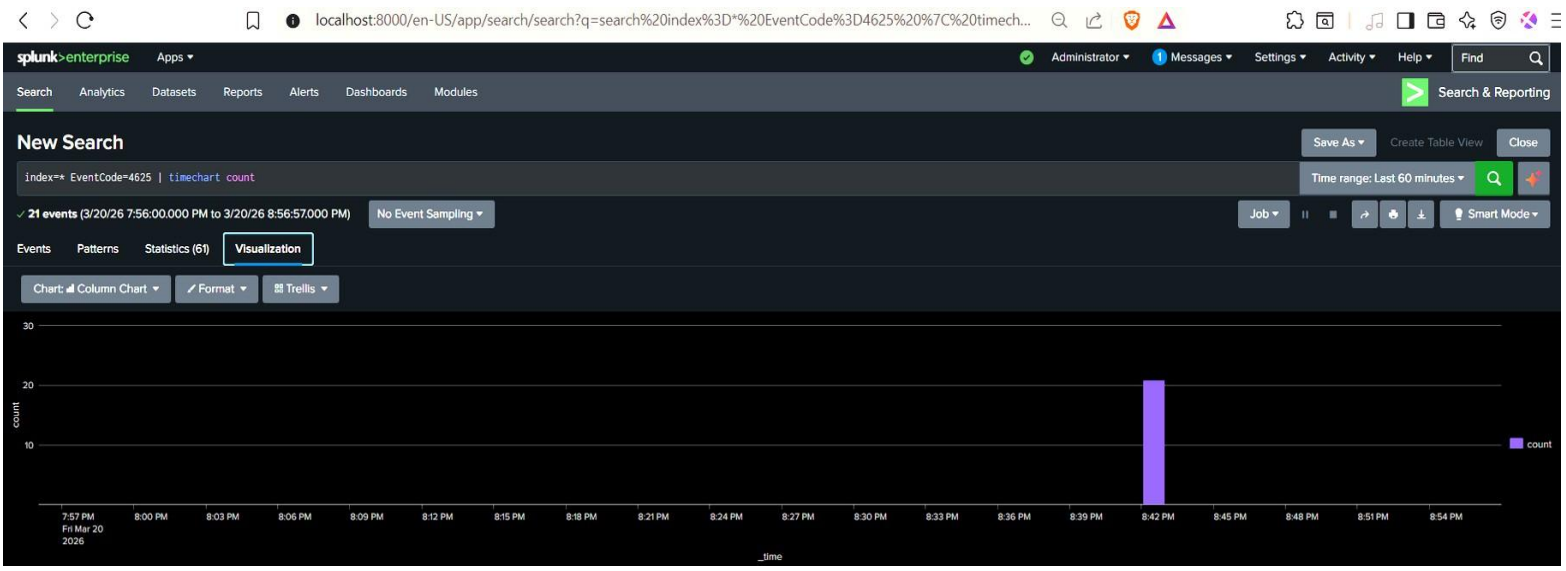
ATTACK VISUALIZATION

Visual analysis of login activity showing abnormal patterns and attack behavior.

Failed Logins Over Time



Top Attacker IP



Attack Spike Visualization